

3379. Transformed Array

Solved 

Easy

 Topics

 Companies

 Hint

You are given an integer array `nums` that represents a circular array. Your task is to create a new array `result` of the **same** size, following these rules:

For each index `i` (where $0 \leq i < \text{nums.length}$), perform the following **independent** actions:

- If `nums[i] > 0`: Start at index `i` and move `nums[i]` steps to the **right** in the circular array. Set `result[i]` to the value of the index where you land.
- If `nums[i] < 0`: Start at index `i` and move `abs(nums[i])` steps to the **left** in the circular array. Set `result[i]` to the value of the index where you land.
- If `nums[i] == 0`: Set `result[i]` to `nums[i]`.

Return the new array `result`.

Note: Since `nums` is circular, moving past the last element wraps around to the beginning, and moving before the first element wraps back to the end.

Example 1:

Input: `nums = [3, -2, 1, 1]`

Output: `[1, 1, 1, 3]`

Explanation:

- For `nums[0]` that is equal to 3, If we move 3 steps to right, we reach `nums[3]`. So `result[0]` should be 1.
- For `nums[1]` that is equal to -2, If we move 2 steps to left, we reach `nums[3]`. So `result[1]` should be 1.
- For `nums[2]` that is equal to 1, If we move 1 step to right, we reach `nums[3]`. So `result[2]` should be 1.
- For `nums[3]` that is equal to 1, If we move 1 step to right, we reach `nums[0]`. So `result[3]` should be 3.

Example 2:

Input: `nums = [-1, 4, -1]`

Output: `[-1, -1, 4]`

Explanation:

- For `nums[0]` that is equal to -1, If we move 1 step to left, we reach `nums[2]`. So `result[0]` should be -1.
- For `nums[1]` that is equal to 4, If we move 4 steps to right, we reach `nums[2]`. So `result[1]` should be -1.
- For `nums[2]` that is equal to -1, If we move 1 step to left, we reach `nums[1]`. So `result[2]` should be 4.

Constraints:

- `1 <= nums.length <= 100`
- `-100 <= nums[i] <= 100`

Python:

class Solution:

```
def constructTransformedArray(self, A: List[int]) -> List[int]:  
    return [A[(i + v) % len(A)] for i, v in enumerate(A)]
```

JavaScript:

```
const constructTransformedArray = A =>  
    A.map((v, i) => A.at((i + v) % A.length));
```

Java:

class Solution:

```
def constructTransformedArray(self, A: List[int]) -> List[int]:  
    return [A[(i + v) % len(A)] for i, v in enumerate(A)]
```